

Kubernetes Assignment-4

Kubernetes Service:

Step 1: Start Minikube

Start the Kubernetes cluster using Docker:

minikube start --driver=docker

Check the Minikube status:

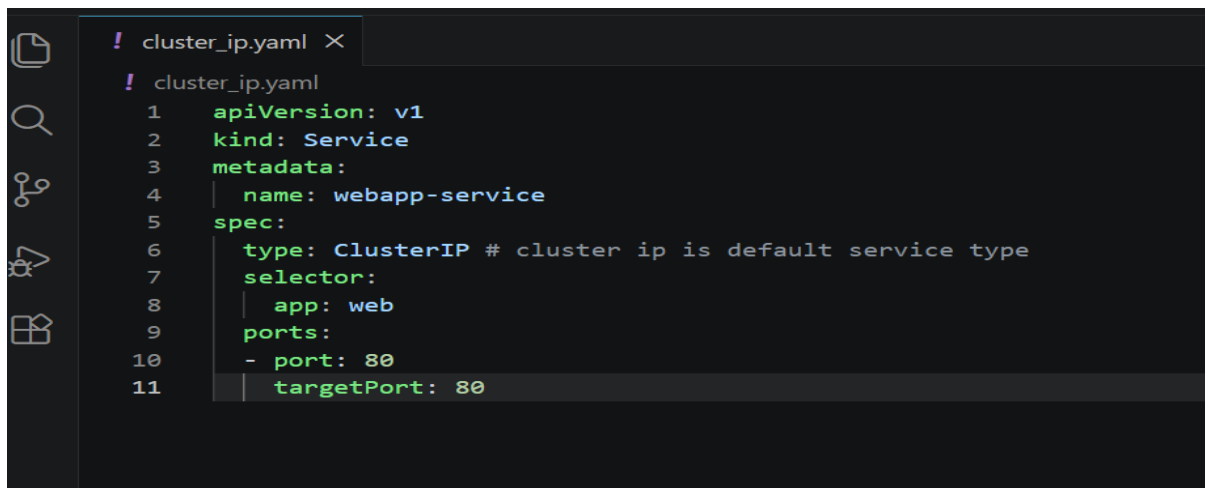
minikube status

Step 2: Check the Kubernetes Node

kubectl get nodes

Step 3: Create the Service YAML File

Create a file using command: **vim cluster_ip.yaml**

A screenshot of a code editor with a dark theme. The editor has a sidebar on the left with icons for file explorer, search, and other functions. The main area shows a file named 'cluster_ip.yaml' with the following content:

```
! cluster_ip.yaml X
! cluster_ip.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: webapp-service
5  spec:
6    type: ClusterIP # cluster ip is default service type
7    selector:
8      app: web
9    ports:
10     - port: 80
11     targetPort: 80
```

Step 4: Create the Service

Apply the YAML file:

kubectl apply -f clusterip.yaml

Step 5: Verify the Service

kubectl get services

Step 6: Check Service Endpoints

kubectl get endpoints webapp-service

Step 6: Test the Service

Create a temporary test Pod:

kubectl run test-pod --image=busybox --restart=Never -- sleep 300

```
File Edit Selection View Go Run ... Kubernetes Assignment-4 Update
! cluster_ip.yaml x
! cluster_ip.yaml
1 apiVersion: v1

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ minikube status
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
app-deployment-c77955c54-pl4rc      1/1      Running   0           9d
app-deployment-c77955c54-plvbx      1/1      Running   0           9d
app-deployment-c77955c54-vf82d      1/1      Running   0           9d
mynginx                              1/1      Running   2 (9d ago)  13d
replicademo-rs-47zdt                1/1      Running   2 (9d ago)  13d
replicademo-rs-98pmj                1/1      Running   2 (9d ago)  13d
replicademo-rs-k92rb                1/1      Running   2 (9d ago)  13d
replicademo-rs-nzvvg                1/1      Running   2 (9d ago)  13d

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl apply -f cluster_ip.yaml
service/webapp-service created

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get services
NAME      TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
kubernetes ClusterIP   10.96.0.1      <none>       443/TCP    13d
webapp-service ClusterIP  10.103.64.15   <none>       80/TCP     12s
```

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get endpoints webapp-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS                                                                                                     AGE
webapp-service 10.244.0.25:80,10.244.0.26:80,10.244.0.29:80 + 4 more... 20m

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get pods --show-labels
NAME                                READY    STATUS    RESTARTS   AGE    LABELS
app-deployment-c77955c54-pl4rc      1/1      Running   0           9d    app=web,pod-template-hash=c77955c54
app-deployment-c77955c54-plvbx      1/1      Running   0           9d    app=web,pod-template-hash=c77955c54
app-deployment-c77955c54-vf82d      1/1      Running   0           9d    app=web,pod-template-hash=c77955c54
mynginx                              1/1      Running   2 (9d ago)  13d    run=mynginx
replicademo-rs-47zdt                1/1      Running   2 (9d ago)  13d    app=web
replicademo-rs-98pmj                1/1      Running   2 (9d ago)  13d    app=web
replicademo-rs-k92rb                1/1      Running   2 (9d ago)  13d    app=web
replicademo-rs-nzvvg                1/1      Running   2 (9d ago)  13d    app=web
```

```
File Edit Selection View Go Run ... Kubernetes Assignment-4 Update
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl run test-pod --image=busybox --restart=Never -- sleep 300
pod/test-pod created

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get pod test-pod
NAME      READY    STATUS    RESTARTS   AGE
test-pod  1/1      Running   0           15s

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl exec test-pod -- wget -qO- http://webapp-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color: scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
```

```
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
```

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes           ClusterIP   10.96.0.1     <none>         443/TCP    13d
webapp-service       ClusterIP   10.103.64.15  <none>         80/TCP     53m

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl port-forward svc/webapp-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```

localhost:8080

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

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To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to fs.com/nginx.

Thank you for using nginx.

Steps to Create a NodePort Service

Step 1: Create the nodeport service YAML File

Create a file using command: **vim nodeport_service.yaml**

```
! nodeport_service.yaml X
! nodeport_service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: myapp-service
5  spec:
6    type: NodePort
7    selector:
8      app: web
9    ports:
10     - port: 80
11       targetPort: 80
12       nodePort: 30380
```

Step 2: Create the Service

Apply the YAML file:

kubectl apply -f nodeport_service.yaml

Step 3: Verify the Service

kubectl get services

```
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1      <none>          443/TCP          13d
myapp-service        NodePort    10.99.125.27   <none>          80:30380/TCP     23m
webapp-service       ClusterIP   10.103.64.15   <none>          80/TCP           145m
```

Step 4: Access it with Minikube

minikube service web-nodeport --url

```
File Edit Selection View Go Run ... Kubernetes Assignment-4 Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl apply -f nodeport_service.yaml
service/myapp-service configured

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get svc myapp-service -o yaml
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"annotations":{},"name":"myapp-service","namespace":"default"},"spec":{"ports":[{"nodePort":30380,"port":80,"targetPort":80}],"selector":{"app":"web"},"type":"NodePort"}}
  creationTimestamp: "2026-08-16T16:35:35Z"
  name: myapp-service
  namespace: default
  resourceVersion: "25807"
  uid: 956ed7ca-80e2-4891-9730-773982a432b9
spec:
  clusterIP: 10.99.125.27
  clusterIPs:
    - 10.99.125.27
  externalTrafficPolicy: Cluster
  internalTrafficPolicy: Cluster
  ipFamilies:
    - IPv4
  ipFamilyPolicy: SingleStack
  ports:
    - nodePort: 30380
      port: 80
      protocol: TCP
      targetPort: 80
  selector:
```

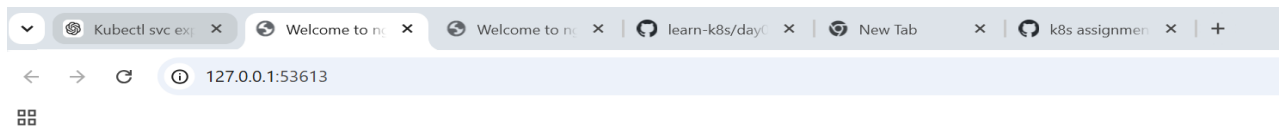
```
File Edit Selection View Go Run ... Kubernetes Assignment-4 Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get svc myapp-service -o yaml
selector:
  app: web
sessionAffinity: None
type: NodePort
status:
  loadBalancer: {}

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get pods --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   LABELS
app-deployment-c77955c54-pl4rc      1/1     Running   0           9d    app=web,pod-template-hash=c77955c54
app-deployment-c77955c54-plvbx      1/1     Running   0           9d    app=web,pod-template-hash=c77955c54
app-deployment-c77955c54-vf82d      1/1     Running   0           9d    app=web,pod-template-hash=c77955c54
mynginx                              1/1     Running   2 (9d ago)  13d   run=mynginx
replicademo-rs-47zdt                1/1     Running   2 (9d ago)  13d   app=web
replicademo-rs-98pmj                1/1     Running   2 (9d ago)  13d   app=web
replicademo-rs-k92rb                1/1     Running   2 (9d ago)  13d   app=web
replicademo-rs-nzvvg                1/1     Running   2 (9d ago)  13d   app=web
test-pod                            0/1     Completed 0           88m   run=test-pod

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$
```

```
File Edit Selection View Go Run ... Kubernetes Assignment-4 Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ kubectl get pods -l app=web
NAME                                READY   STATUS    RESTARTS   AGE   LABELS
app-deployment-c77955c54-pl4rc      1/1     Running   0           9d    app=web,pod-template-hash=c77955c54
app-deployment-c77955c54-plvbx      1/1     Running   0           9d    app=web,pod-template-hash=c77955c54
app-deployment-c77955c54-vf82d      1/1     Running   0           9d    app=web,pod-template-hash=c77955c54
replicademo-rs-47zdt                1/1     Running   2 (9d ago)  13d   app=web
replicademo-rs-98pmj                1/1     Running   2 (9d ago)  13d   app=web
replicademo-rs-k92rb                1/1     Running   2 (9d ago)  13d   app=web
replicademo-rs-nzvvg                1/1     Running   2 (9d ago)  13d   app=web

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-4 (main)
$ minikube service myapp-service --url
http://127.0.0.1:53613
Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



Welcome to nginx!

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